

Algebra 1
Unit 11
Lesson 5 Practice

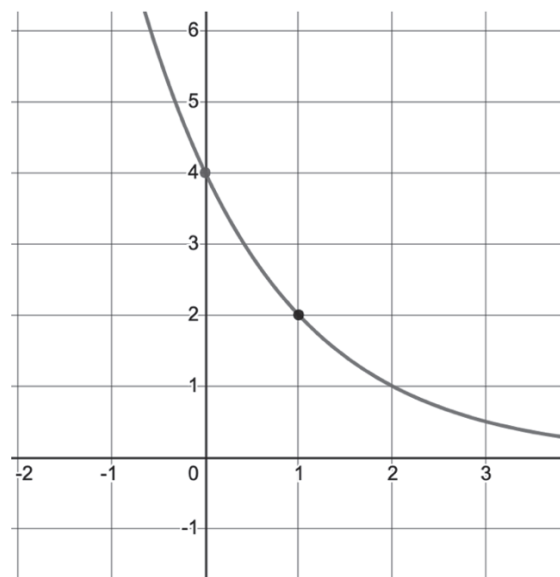
Name _____
Date _____
Period _____

The graph of an exponential function is shown.

1. Find the value of a by substituting the point $(0, 4)$ in for x and y in $y = a(b)^x$.

$$\boxed{} = a(b)^{\boxed{}}$$

2. Use the value of a found in question 1 and the point $(1, 2)$ to determine the value of b .



3. Write the function for the graph in both forms.

$f(x) = a(b)^x$	$f(x) = a(1 \pm r)^x$

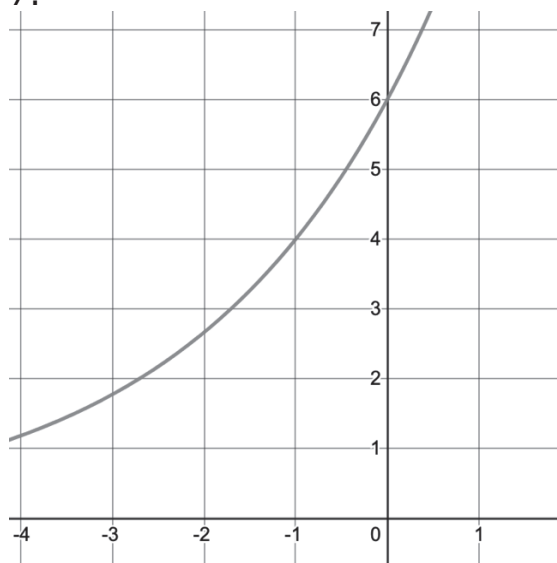
The table of an exponential function is shown.

x	0	1	2	3	4
$f(x)$	0.4	1.2	3.6	10.8	32.4

4. What is the initial value, a ?
5. How does the value of $f(x)$ change as the x -value increases by 1?
6. Write the function in the form $f(x) = a(b)^x$

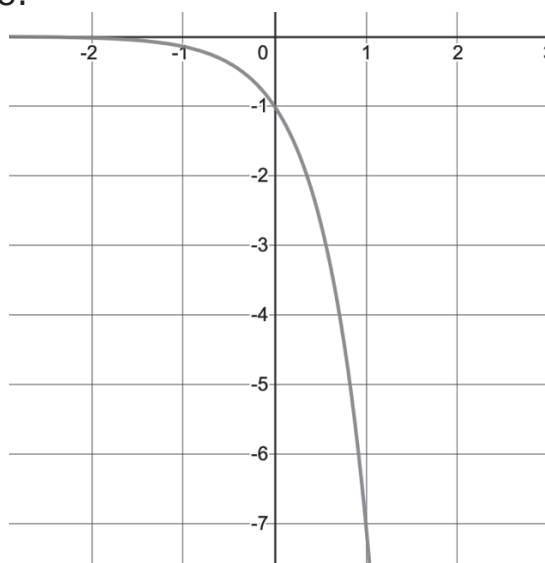
For questions 7-10, write an equation from the table or graph by finding the initial value, a , and common ratio, b .

7.



Function

8.



Function

9.

x	y
0	-0.5
1	-4
2	-32
3	-256
4	-2048

Function

10.

x	y
0	3
1	$\frac{15}{4}$
2	$\frac{75}{16}$
3	$\frac{375}{64}$
4	$\frac{1875}{256}$

Function